DEVELOPMENT OF AN

Ethernet-Based Vehicle Monitoring System

Using Linux Socket Programming in C

Transport Protocol Behaviour for Vehicle Monitoring

Project platform: Pop!_OS / Linux

Communication: TCP and UDP sockets

1. Overview

TCP and UDP are transport-layer protocols that applications can use to exchange vehicle monitoring data. The choice depends on delivery guarantees, latency requirements, and application design.

2. Comparative Table

Parameter	TCP	UDP
Reliability	Acknowledgements, retransmission, and ordered byte-stream delivery.	No built-in delivery or ordering guarantee.
Connection establishment	Connection-oriented; normally uses a three-way handshake.	Connectionless; datagrams can be sent without a handshake.
Overhead	Larger transport header and connection state; may include ACKs and retransmissions.	Smaller base header and no connection state in the transport protocol.
Error handling	Detects segment corruption and retransmits lost data as needed.	Checksum detects corruption where used; delivery recovery is application-specific.
Data model	Byte stream; application must frame messages.	Datagrams preserve message boundaries.
Latency characteristics	Retransmission and congestion control can add delay.	Lower protocol overhead, but delivery is not guaranteed.
Typical usage	Web sessions, file transfer, commands that require reliable delivery.	Telemetry, discovery, real-time updates where occasional loss is acceptable.

3. Application to Vehicle Monitoring

In this demonstration, UDP sends periodic speed, RPM, and steering readings to port 5005. It is simple and has low protocol overhead, but a lost datagram is not automatically retransmitted.

The TCP implementation sends the same readings over a connection to port 5006. TCP provides ordered reliable delivery while the connection remains viable, but it does not preserve application message boundaries. The newline delimiter in the demonstration provides message framing.

4. Interpreting Wireshark Captures

- UDP: inspect datagram ports, length, and the application payload.
- TCP: identify SYN, SYN-ACK, ACK, subsequent data, and FIN/ACK teardown.
- A single application message may span multiple TCP segments, or several messages may be carried in a segment.
- Loopback captures are suitable for demonstrating socket behaviour, but they do not measure physical Ethernet performance.

5. Conclusion

TCP is appropriate when reliable ordered delivery is required. UDP is useful for compact, low-overhead datagrams when the application can tolerate loss or implements its own recovery. The correct choice depends on the vehicle system's functional and timing requirements.